SESSION 3

WHERE Clause & Operators

Complete Original Questions • SQL Answers • Easy English Explanations

Session Topics

- WHERE operators: =, <>, >, <, <=, >=
- Logical operators: AND, OR, NOT

Demo: Filter employees with salary > 50000 and in "IT" dept.

Learning goal: Use WHERE to filter rows and use comparison and logical operators to match the conditions in each task.

Task 1

Question: Write an SQL query to select all restaurants from a table named 'restaurants' where the rating is greater than or equal to 4.5.

Answer

```
SELECT *
FROM restaurants
WHERE rating >= 4.5;
```

Explanation

- SELECT * displays all columns for each restaurant that matches the condition.
- FROM restaurants tells SQL to read the data from the restaurants table.
- WHERE rating >= 4.5 keeps restaurants whose rating is 4.5 or higher.
- The >= operator means “greater than or equal to”, so a rating of exactly 4.5 is included.

Result: Only restaurants with ratings of 4.5 or above appear in the output.

Task 2

Question: In a table called 'movies', filter and display only the movies released after 2020 and with genre 'Action' using the WHERE clause and AND operator.

Answer

```
SELECT *
FROM movies
WHERE release_year > 2020
  AND genre = 'Action';
```

Explanation

- SELECT * displays all columns of the matching movie records.

- FROM movies tells SQL to read data from the movies table.
- release_year > 2020 selects movies released after 2020. Movies released in 2020 are not included.
- genre = 'Action' selects rows where the genre is Action.
- AND means both conditions must be true: the movie must be released after 2020 and have the Action genre.

Result: Only movies released in 2021 or later whose genre is Action are displayed.

Task 3

Question: Given a table 'products' with columns (id, name, price, category), write a query to find all products not in the 'Electronics' category or with a price less than 500.

Answer

```
SELECT *
FROM products
WHERE category <> 'Electronics'
   OR price < 500;
```

Explanation

- SELECT * displays every column for the products that match the condition.
- FROM products reads the records from the products table.
- category <> 'Electronics' selects products whose category is not Electronics. The <> operator means “not equal to”.
- price < 500 selects products whose price is strictly below 500. A price of exactly 500 does not satisfy this condition.
- OR means at least one condition must be true. A product is included if it is not in Electronics, if its price is below 500, or if both are true.
- Example: an Electronics product priced at 450 is included because price < 500 is true. An Electronics product priced at 700 is excluded because both conditions are false.

Important: The question says OR. Using AND instead would change the meaning because both conditions would then need to be true.

Task 4

Question: Write an SQL query for a table 'users' to show all users who are NOT from 'Ahmedabad' and have more than 1000 followers.

Hint from the task: Use the NOT operator combined with AND.

Answer

```
SELECT *
FROM users
WHERE NOT (city = 'Ahmedabad')
  AND followers > 1000;
```

Explanation

- SELECT * displays all columns for the users who match both conditions.
- FROM users reads data from the users table.
- NOT (city = 'Ahmedabad') excludes users whose city is Ahmedabad.
- followers > 1000 selects users with more than 1000 followers. A user with exactly 1000 followers is not included.
- AND means both conditions must be true: the user must not be from Ahmedabad and must have more than 1000 followers.
- Parentheses make it clear that NOT applies to the city comparison.

Assumption: The users table contains the city and followers columns, as implied by the task.

Quick Revision

Comparison operators: = means equal to; <> means not equal to; > means greater than; < means less than; >= means greater than or equal to; <= means less than or equal to.

Logical operators: AND requires all conditions to be true; OR requires at least one condition to be true; NOT reverses a condition.

WHERE clause: Filters rows before they appear in the query result.